Recursive PAC-Bayes

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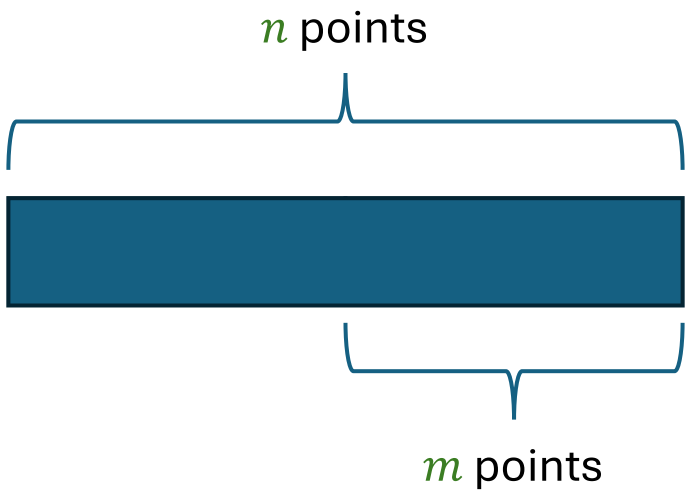
PAC-Bayes reminder

- Theorem (PAC-Bayes-kl): For any "prior" distribution π on $\mathcal{H}$ that is independent of S

$$\mathbb{P}\left(\exists\rho: \mathrm{kl}\left(\mathbb{E}_\rho\left[\hat{L}(h,S)\right] \| \mathbb{E}_\rho[L(h)]\right) \geq \frac{\mathrm{KL}(\rho\|\pi) + \ln\frac{2\sqrt{n}}{\delta}}{n}\right) \leq \delta$$

- How to construct a "good" π?

Data-Informed Priors



- $\mathbb{P}\left(\exists\rho: \mathrm{kl}\left(\mathbb{E}_\rho\left[\hat{L}(h,S)\right] || \mathbb{E}_\rho[L(h)]\right) \geq \frac{\mathrm{KL}(\rho||\pi)+\ln\frac{2\sqrt{n}}{\delta}}{n}\right) \leq \delta$

- Idea: use part of the data to construct (data-informed) π and the rest of the data to compute the bound

- Advantage: “good” π

- Disadvantage: the denominator of the bound decreases from n to m

- Challenge: the confidence information on the prior is lost
  - How much data were used to construct the prior?
  - Was it constructed in a single step or multiple steps?
  - Multi-step processing is not very helpful

Recursive PAC-Bayes

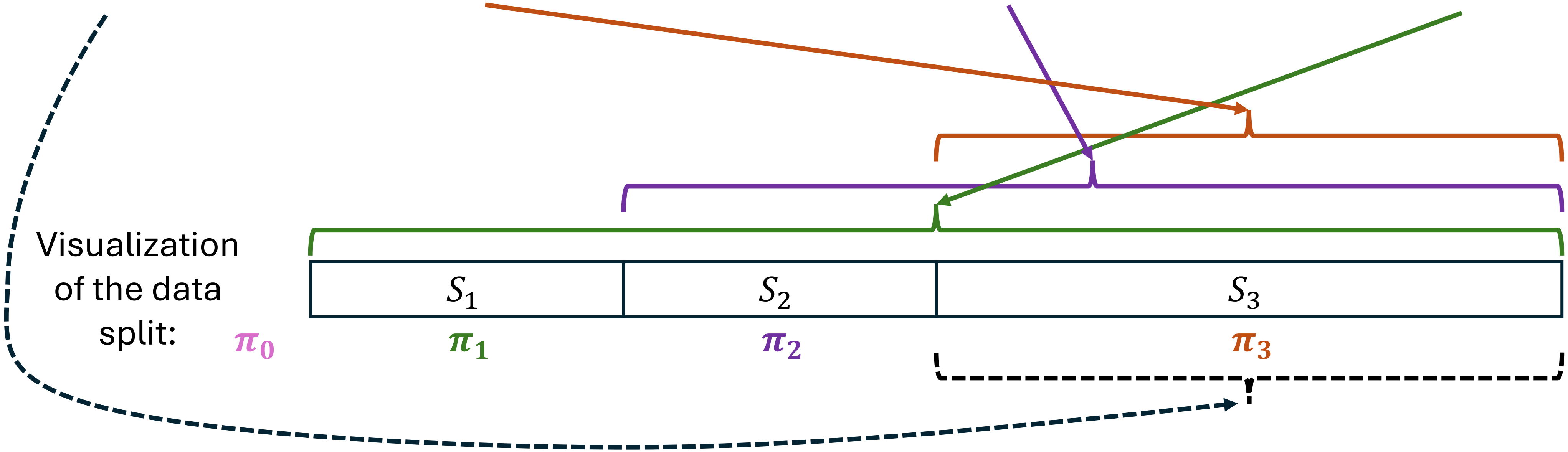
- Recursive loss decomposition

$$\mathbb{E}_{\pi_t}[L(h)] = \mathbb{E}_{\pi_t}\Big[\underbrace{L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]}_{\text{Excess loss (small variance)}}\Big] + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}}[L(h')]}_{\text{Decompose recursively}}$$

$$\mathbb{E}_{\pi_t}[L(h)] = \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]\right] + \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]$$

Illustration: 3-terms Recursive Decomposition

$$\mathbb{E}_{\pi_3}[L(h)] = \mathbb{E}_{\pi_3}\left[L(h) - \gamma_3 \mathbb{E}_{\pi_2}[L(h')]\right] + \gamma_3 \overbrace{\left(\mathbb{E}_{\pi_2}\left[L(h) - \gamma_2 \mathbb{E}_{\pi_1}[L(h')]\right] + \gamma_2 \mathbb{E}_{\pi_1}[L(h)]\right)}^{\mathbb{E}_{\pi_2}[L(h)]}$$



$$\mathbb{E}_{\pi_t}[L(h)] = \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]\right] + \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]$$

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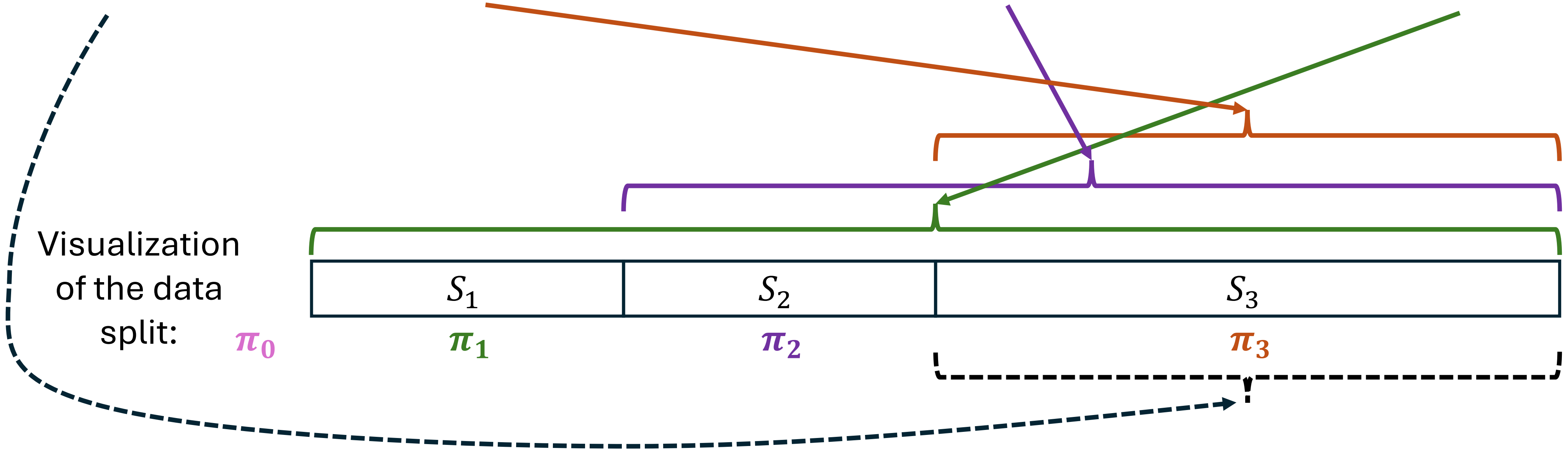
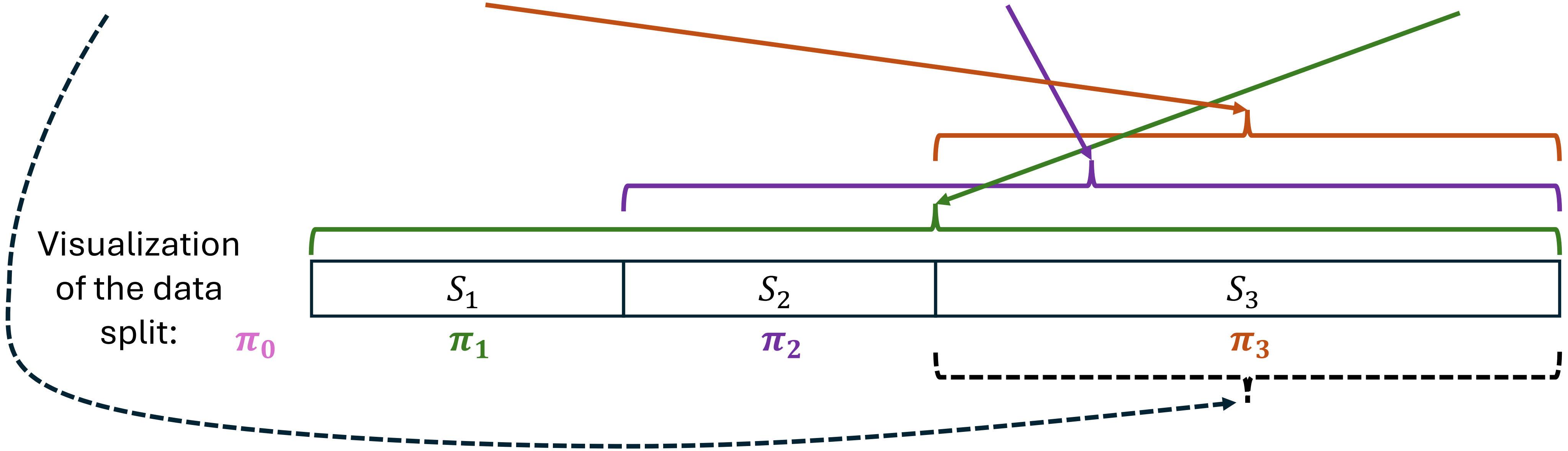


- Early steps:
  - Poor $\pi_{t-1} \Rightarrow$ large $\mathrm{KL}(\pi_t || \pi_{t-1})$
  - Large denominator
  - Small $\gamma_t \cdot \ldots \cdot \gamma_T$
- Late steps:
  - Good $\pi_{t-1} \Rightarrow$ small $\mathrm{KL}(\pi_t || \pi_{t-1})$
  - Benefit from small variance of the excess loss

$$\mathbb{E}_{\pi_t}[L(h)] = \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]\right] + \gamma_t \mathbb{E}_{\pi_{t-1}}[L(h')]$$

Illustration: 3-terms Recursive Decomposition

$$\mathbb{E}_{\pi_3}[L(h)] = \mathbb{E}_{\pi_3}\left[L(h) - \gamma_3 \mathbb{E}_{\pi_2}[L(h')]\right] + \gamma_3 \overbrace{\left(\mathbb{E}_{\pi_2}\left[L(h) - \gamma_2 \mathbb{E}_{\pi_1}[L(h')]\right] + \gamma_2 \mathbb{E}_{\pi_1}[L(h)]\right)}^{\mathbb{E}_{\pi_2}[L(h)]}$$



- A good way to split the data – geometric split ($|S_t| = 2|S_{t-1}|$)
  - Use little data to bring π_t to a "good region"
  - Leave sufficient data to bound the last term $\left(|S_T| = \frac{1}{2}|S|\right)$

Recursive PAC-Bayes bound

$$\underbrace{\mathbb{E}_{\pi_t}[L(h)]}_{\leq B_t(\pi_t)} = \underbrace{\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right]}_{\leq \mathcal{E}_t(\pi_t, \gamma_t)} + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}^*}[L(h')]}_{\leq B_{t-1}(\pi_{t-1}^*)}$$

$$B_t(\pi_t) = \mathcal{E}_t(\pi_t, \gamma_t) + B_{t-1}(\pi_{t-1}^*)$$

- Theorem: Let $\pi_0^*, \pi_1^*, \ldots, \pi_T^*$ be such that π_t^* is independent of $\bigcup_{s=t}^T S_t$.

  Assume there exists $B_1(\pi_1)$ such that

  $$\mathbb{P}\left(\exists \pi_1 : \mathbb{E}_{\pi_1}[L(h)] \geq B_1(\pi_1)\right) \leq \frac{\delta}{T}.$$

  Assume there exists $\mathcal{E}_t(\pi_t, \gamma_t)$ such that

  $$\mathbb{P}\left(\exists \pi_t : \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right] \geq \mathcal{E}_t(\pi_t, \gamma_t)\right) \leq \frac{\delta}{T}.$$

  Let

  $$B_t(\pi_t) = \mathcal{E}_t(\pi_t, \gamma_t) + \gamma_t B_{t-1}(\pi_{t-1}^*).$$

  Then

  $$\mathbb{P}\left(\exists t \in \{1, \ldots, T\} \; and \; \pi_t : \mathbb{E}_{\pi_t}[L(h)] \geq B_t(\pi_t)\right) \leq \delta.$$

The Excess Loss $\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right]$

$$\begin{aligned}
\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right] &= \mathbb{E}_{\pi_t}\left[\mathbb{E}_{\pi_{t-1}^*}[L(h) - \gamma_t L(h')]\right] \\
&= \mathbb{E}_{\pi_t}\left[\mathbb{E}_{\pi_{t-1}^*}[\mathbb{E}_{\mathcal{D}}[\ell(h(X),Y) - \gamma_t \ell(h'(X),Y)]]\right] \\
&= \mathbb{E}_{\pi_t}\left[\mathbb{E}_{\pi_{t-1}^* \times \mathcal{D}}[\ell(h(X),Y) - \gamma_t \ell(h'(X),Y)]\right]
\end{aligned}$$

$$f_{\gamma_t}\big(h,(h',X,Y)\big) = \ell(h(X),Y) - \gamma_t \ell(h'(X),Y) \in \{-\gamma_t, 0, 1-\gamma_t, 1\}$$

[Previously $\ell\big(h,(X,Y)\big) = \ell(h(X),Y)$]

$$F_{\gamma_t}(h) = \mathbb{E}_{\pi_{t-1}^* \times \mathcal{D}}\left[f_{\gamma_t}\big(h,(h',X,Y)\big)\right] = L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]$$

$$\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right] = \mathbb{E}_{\pi_t}[F_{\gamma_t}(h)]$$

Bounding $\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi^*_{t-1}}[L(h')]\right]$

$$f_{\gamma_t}\big(h, (h', X, Y)\big) = \ell(h(X), Y) - \gamma_t \ell(h'(X), Y) \in \{-\gamma_t, 0, 1 - \gamma_t, 1\}$$

$$\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi^*_{t-1}}[L(h')]\right] = \mathbb{E}_{\pi_t}\left[F_{\gamma_t}(h)\right] = \mathbb{E}_{\pi_t}\left[\mathbb{E}_{\pi^*_{t-1} \times \mathcal{D}}\left[f_{\gamma_t}\big(h, (h', X, Y)\big)\right]\right]$$

1. Generate a sample of $F_{\gamma_t}(h)$: for each $(X_i, Y_i) \in S$ sample $h'_i \sim \pi^*_{t-1}$ to create (h'_i, X_i, Y_i).
2. f_{γ_t} takes 4 values $\{-\gamma_t, 0, 1 - \gamma_t, 1\} = \{b_0, b_1, b_2, b_3\}$. Use the same binary decomposition as in split-kl: $f_{|j}(\cdot,\cdot) = \mathbb{1}\big(f(\cdot,\cdot) \geq b_j\big)$.

PAC-Bayes-split-kl

- Let $f(h,Z) \in \{b_0, \dots, b_K\}$ and $F(h) = \mathbb{E}_Z[f(h,Z)]$.
- Let $f_{|j}(\cdot,\cdot) = \mathbb{1}\big(f(\cdot,\cdot) \geq b_j\big)$ and $\alpha_j = b_j - b_{j-1}$.
  Then $f(\cdot,\cdot) = b_0 + \sum_{j=1}^K \alpha_j f_{|j}(\cdot,\cdot)$
- Let $F_{|j}(h) = \mathbb{E}_Z[f_{|j}(h,Z)]$ and $\hat{F}_{|j}(h,S) = \frac{1}{n}\sum_{i=1}^n f_{|j}(h,Z_i)$

- Theorem: For π independent of S,

$$\mathbb{P}\left(\exists\rho: \mathbb{E}_\rho[F(h)] \geq b_0 + \sum_{j=1}^K \alpha_j \mathrm{kl}^{-1^+}\left(\mathbb{E}_\rho\big[\hat{F}_{|j}(h,S)\big], \frac{\mathrm{KL}(\rho||\pi) + \ln\frac{2K\sqrt{n}}{\delta}}{n}\right)\right) \leq \delta$$

- Proof: A union bound on PAC-Bayes-kl-s on the deviations of $\mathbb{E}_\rho\big[\hat{F}_{|j}(h,S)\big]$ from $\mathbb{E}_\rho\big[F_{|j}(h)\big]$.

Recursive PAC-Bayes-split-kl

$$\underbrace{\mathbb{E}_{\pi_t}[L(h)]}_{\leq B_t(\pi_t)} = \underbrace{\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right]}_{\leq \mathcal{E}_t(\pi_t,\gamma_t)} + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}^*}[L(h')]}_{\leq B_{t-1}(\pi_{t-1}^*)}$$

- For $t = 1$ by PAC-Bayes-kl:

$$\mathbb{P}\left(\exists \pi_1 : \mathbb{E}_{\pi_1}[L(h)] \geq \underbrace{\mathrm{kl}^{-1^+}\left(\mathbb{E}_{\pi_1}\left[\hat{L}(h,S)\right], \frac{\mathrm{KL}(\pi_1 \| \pi_0^*) + \ln\frac{2T\sqrt{n}}{\delta}}{n}\right)}_{B_1(\pi_1)}\right) \leq \frac{\delta}{T}$$

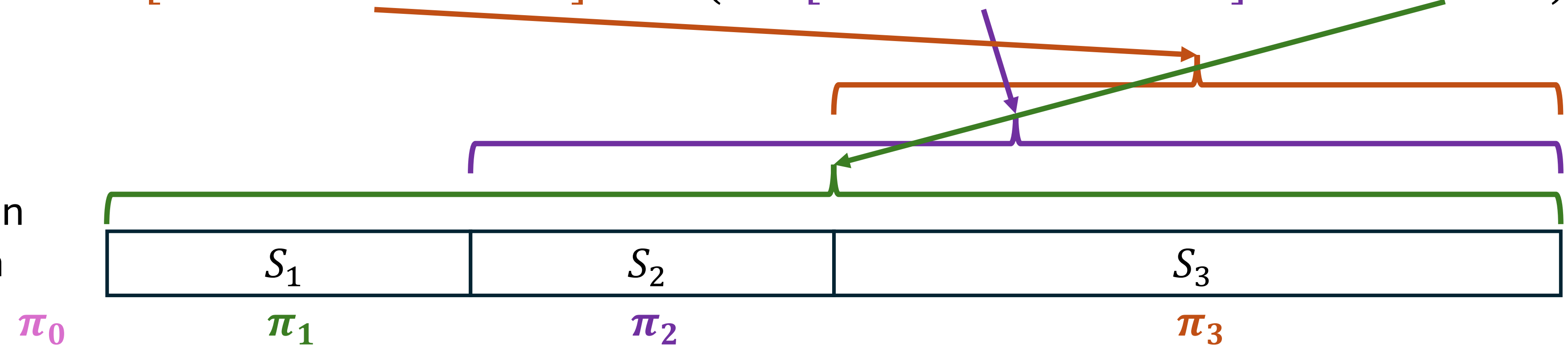
- For $t \geq 2$ by PAC-Bayes-split-kl:

$$\mathbb{P}\left(\exists \pi_t : \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right] \geq \underbrace{-\gamma_t + \sum_{j=1}^{3} \alpha_j \mathrm{kl}^{-1^+}\left(\mathbb{E}_{\pi_t}\left[\hat{F}_{\gamma_t|j}(h,U_t)\right], \frac{\mathrm{KL}(\pi_t \| \pi_{t-1}^*) + \ln\frac{6T\sqrt{m_t}}{\delta}}{m_t}\right)}_{\mathcal{E}_t(\pi_t,\gamma_t)}\right) \leq \frac{\delta}{T}$$

Getting the samples right

$$\mathbb{E}_{\pi_3}[L(h)] = \mathbb{E}_{\pi_3}\left[L(h) - \gamma_3 \mathbb{E}_{\pi_2}[L(h')]\right] + \gamma_3 \overbrace{\left(\mathbb{E}_{\pi_2}\left[L(h) - \gamma_2 \mathbb{E}_{\pi_1}[L(h')]\right] + \gamma_2 \mathbb{E}_{\pi_1}[L(h)]\right)}^{\mathbb{E}_{\pi_2}[L(h)]}$$

Visualization of the data split:



- $\mathbb{P}\left(\exists \pi_t : \mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi^*_{t-1}}[L(h')]\right] \geq \underbrace{-\gamma_t + \sum_{j=1}^{3} \alpha_j \mathrm{kl}^{-1^+}\left(\mathbb{E}_{\pi_t}[\hat{F}_{\gamma_t|j}(h, U_t)], \frac{\mathrm{KL}(\pi_t\|\pi^*_{t-1}) + \ln\frac{6T\sqrt{m_t}}{\delta}}{m_t}\right)}_{\mathcal{E}_t(\pi_t, \gamma_t)}\right) \leq \frac{\delta}{T}$
- In construction of π^*_t : $U_t = \hat{\pi}^*_{t-1} \circ S_t$
- In calculation of the final bound: $U_t = \hat{\pi}^*_{t-1} \circ \bigcup_{s=t}^{T} S_s$ and $m_t = |\bigcup_{s=t}^{T} S_s|$
- In construction of π^*_t we can still take $m_t = |\bigcup_{s=t}^{T} S_s| \geq |S_t|$, because we know we will have more data later
  - Allows more aggressive deviation of π_t from π_{t-1}
  - $\mathbb{E}_{\pi_t}[\hat{F}_{\gamma_t}(h, \hat{\pi}^*_{t-1} \circ S_t)] \leq \mathbb{E}_{\pi_t}[\hat{F}_{\gamma_t}(h, \hat{\pi}^*_{t-1} \circ \bigcup_{s=t}^{T} S_s)]$, so the effect on the final bound depends on the data

Selection of γ_t

$$\underbrace{\mathbb{E}_{\pi_t}[L(h)]}_{\leq B_t(\pi_t)} = \underbrace{\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right]}_{\leq \mathcal{E}_t(\pi_t, \gamma_t)} + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}^*}[L(h')]}_{\leq B_{t-1}(\pi_{t-1}^*)}$$

- We want $B_t(\pi_t^*) = \mathcal{E}_t(\pi_t^*, \gamma_t) + \gamma_t B_t(\pi_{t-1}^*) < B_{t-1}(\pi_{t-1}^*)$.
- $\gamma_t < 1 - \frac{\mathcal{E}_t(\pi_t, \gamma_t)}{B_{t-1}(\pi_{t-1}^*)} < 1$
- And $\gamma_t \geq 0$
- $\mathcal{E}_t(\pi_t, \gamma_t)$ decreases with γ_t, whereas $\gamma_t B_t(\pi_{t-1}^*)$ increases with γ_t
- The optimal choice depends on the data
- γ_t should be independent of $S_t, \dots, S_T$, but can depend on $S_1, \dots, S_{t-1}$
- We can take a grid of γ_t and the best value from the grid at the cost of a union bound
- In experiments we simply took $\gamma_t = \frac{1}{2}$. Further experimentation is welcome.

How to split the sample

- Geometric split works well.
  - $|S_T| = \frac{1}{2}|S|$
  - $|S_t| = \frac{1}{2}|S_{t+1}|$

- Little data are used to bring π_t to a "good" region
- Leaves sufficient data for building $\mathcal{E}_T(\pi_T, \gamma_T)$

Empirical evaluation

	MNIST			Fashion MNIST		
	Train 0-1	Test 0-1	Bound	Train 0-1	Test 0-1	Bound
Uninf.	.343 (2e-3)	.335 (3e-3)	.457 (2e-3)	.382 (2e-3)	.384 (2e-3)	.464 (2e-3)
Inf.	.377 (8e-4)	.371 (6e-3)	.408 (9e-4)	.412 (1e-3)	.413 (6e-3)	.440 (1e-3)
Inf. + Ex.	.157 (2e-3)	.151 (3e-3)	.192 (2e-3)	.280 (4e-3)	.285 (5e-3)	.342 (6e-3)
RPB $T = 2$	.143 (2e-3)	.139 (3e-3)	.321 (3e-3)	.257 (3e-3)	.266 (5e-3)	.404 (3e-3)
RPB $T = 4$	.112 (1e-3)	.109 (1e-3)	.203 (8e-4)	.203 (2e-3)	.213 (3e-3)	.293 (1e-3)
RPB $T = 6$	.103 (1e-3)	.101 (1e-3)	.166 (1e-3)	.186 (4e-4)	.198 (1e-3)	.255 (1e-3)
RPB $T = 8$	**.101 (1e-3)**	**.097 (2e-3)**	**.158 (2e-3)**	**.181 (1e-3)**	**.192 (3e-3)**	**.242 (1e-3)**

- Uninformed π
- Informed π using half of the data
- Informed π with excess loss
  - Mhammedi et al. (2019)
  - $\mathbb{E}_\rho[L(h)] = \mathbb{E}_\rho[L(h) - L(h^*)] + L(h^*)$
  - π and h^* trained on half of the data
- Recursive PAC-Bayes (RPB) with $S = S_1, \ldots, S_T$

Summary

- $\underbrace{\mathbb{E}_{\pi_t}[L(h)]}_{\le B_t(\pi_t)} = \underbrace{\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t \mathbb{E}_{\pi_{t-1}^*}[L(h')]\right]}_{\le \mathcal{E}_t(\pi_t, \gamma_t)} + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}^*}[L(h')]}_{\le B_{t-1}(\pi_{t-1}^*)}$
- $B_t(\pi_t) = \mathcal{E}_t(\pi_t, \gamma_t) + \gamma_t B_{t-1}(\pi_{t-1}^*)$
- Use any “plain” PAC-Bayes bound (e.g., PAC-Bayes-kl) to bound $B_1(\pi_1^*)$
- Use any “excess loss” bound (e.g., PAC-Bayes-split-kl) to bound $\mathcal{E}_t(\pi_t, \gamma_t)$
- $\gamma_t B_{t-1}(\pi_{t-1}^*)$ preserves confidence information
- $\mathcal{E}_t(\pi_t, \gamma_t)$ is an “excess loss” (provides tighter bounds)
- Geometric split works well – uses little data to shape π_t and leaves sufficient data for tight $\mathcal{E}_T(\pi_T, \gamma_T)$
- Significant empirical improvements

Open Question

- Derive recursive bounds for the weighted majority vote

- A simple, but inefficient approach:
$$L\left(MV_{\pi_t}\right) \leq 2\mathbb{E}_{\pi_t}[L(h)] = 2\mathbb{E}_{\pi_t}\left[L(h) - \gamma_t L\left(MV_{\pi_{t-1}}\right)\right] + 2\gamma_t L(MV_{\pi_{t-1}})$$
- Factor 2 will get amplified through the recursion
  - Even though $L(MV_{\pi_{t-1}})$ is a reasonable quantity to take as a reference loss!

- Can we relate $L\left(MV_{\pi_t}\right)$ to $L(MV_{\pi_{t-1}})$ more directly?
  - Without going through the known oracle inequalities that have the constant factors
- Or derive better bounds for $L\left(MV_{\pi_t}\right)$?
  - That will not suffer from the constant factors in the current oracle bounds

Bernstein’s Inequalities

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Motivation

- Ways to exploit small variance to guarantee tighter concentration (same as in split-kl, but with other tools)

Bernstein’s Inequality

- Theorem: Let $X_1, \ldots, X_n$ be i.i.d., such that $\mathbb{E}[X_1] - X_1 \leq b$ and $\mathbb{V}[X_1] \leq \nu$. Let $\mu = \mathbb{E}[X_1]$ and $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then

$$\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\nu \ln\frac{1}{\delta}}{n}} + \frac{b \ln\frac{1}{\delta}}{3n}\right) \leq \delta.$$

- “Fast rate” when ν is small

Bernstein's Inequality – Proof idea

- Bernstein's Inequality: $\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\nu \ln\frac{1}{\delta}}{n}} + \frac{b \ln\frac{1}{\delta}}{3n}\right) \leq \delta$

- Bernstein's Lemma: For r.v. $X \leq b$ and any $\lambda \in \left(0, \frac{3}{b}\right)$

$$\mathbb{E}\left[e^{\lambda(\mathbb{E}[X]-X)-\frac{\lambda^2}{2\left(1-\frac{b\lambda}{3}\right)}\mathbb{V}[X]}\right] \leq 1$$

- Chernoff's bounding technique:
  - Take things to the exponent
  - Represent exponent of a sum as a product of exponents
  - Apply independence to swap product and expectation
  - Apply Bernstein's Lemma
  - Find the optimal λ (happens to be independent of $\hat{\mu}_n$)

Empirical Bernstein's inequality

- Bernstein's inequality: $\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\nu \ln\frac{1}{\delta}}{n}} + \frac{b \ln\frac{1}{\delta}}{3n}\right) \leq \delta$

- What if we do not know $\mathbb{V}[X_1] \leq \nu$?

- Theorem: Let $X_1, \dots, X_n$ be i.i.d. in $[0,1]$ $(b = 1)$. Let $\mu = \mathbb{E}[X_1]$, $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i$, and $\hat{\nu}_n = \frac{1}{n(n-1)}\sum_{1\leq i<j\leq n}\left(X_i - X_j\right)^2$. Then for $n \geq 5$

$$\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\hat{\nu}_n \ln\frac{2}{\delta}}{n}} + \frac{7\ln\frac{2}{\delta}}{3n}\right) \leq \delta.$$

- Proof idea: derive a high-probability bound on $\nu - \hat{\nu}_n$ and plug it into Bernstein's inequality.
  - $\ln\frac{2}{\delta}$ - the cost of a union bound

Comparison to earlier bounds

- Assume $X_1, \dots, X_n$ i.i.d. in $[0,1]$.
- Let $\mu = \mathbb{E}[X_1]$, $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i$, $\hat{\nu}_n = \frac{1}{n(n-1)}\sum_{1\le i<j\le n}\left(X_i - X_j\right)^2$.
- Hoeffding: $\mathbb{P}\left(\mu \ge \hat{\mu}_n + \sqrt{\frac{\ln\frac{1}{\delta}}{2n}}\right) \le \delta$
- Refined Pinsker's relaxation of kl:

$$\mathbb{P}\left(\mu \ge \hat{\mu}_n + \sqrt{\frac{2\hat{\mu}_n \ln\frac{1}{\delta}}{n}} + \frac{2\ln\frac{1}{\delta}}{n}\right) \le \delta$$

- Empirical Bernstein:

$$\mathbb{P}\left(\mu \ge \hat{\mu}_n + \sqrt{\frac{2\hat{\nu}_n \ln\frac{1}{\delta}}{n}} + \frac{7\ln\frac{1}{\delta}}{3n}\right) \le \delta$$

- Claim: $\hat{\nu}_n \le \hat{\mu}_n$ and $\hat{\nu}_n \le \frac{n}{n-1}\frac{1}{4}$

- Hoeffding – "zeroth order" – "slow rate"
- kl – "first order" – "fast rate" for small $\hat{\mu}_n$
- Empirical Bernstein – "second order" – "fast rate" for small $\hat{\nu}_n$

- The major advantage of Empirical Bernstein is when $\hat{\nu}_n$ is small, but $\hat{\mu}_n$ is large.
- If $\hat{\mu}_n$ is small, Empirical Bernstein has no advantage.
- kl is still better for Bernoulli
- Comparison to split-kl is more involved

Unexpected Bernstein's Inequality

- Unexpected Bernstein's Lemma: For r.v. $X \leq b$ and any $\lambda \in (0, \frac{1}{b})$
$$\mathbb{E}\left[e^{\lambda(\mathbb{E}[X]-X)+\frac{b\lambda+\ln(1-b\lambda)}{b^2}X^2}\right] \leq 1$$

- Bernstein's Lemma: For r.v. $X \leq b$ and any $\lambda \in \left(0, \frac{3}{b}\right)$
$$\mathbb{E}\left[e^{\lambda(\mathbb{E}[X]-X)-\frac{\lambda^2}{2\left(1-\frac{b\lambda}{3}\right)}\mathbb{V}[X]}\right] \leq 1$$

- Key difference: relates $\mathbb{E}[X] - X$ to X^2 (empirical quantity) rather than $\mathbb{V}[X]$ (oracle quantity)

Unexpected Bernstein's Inequality

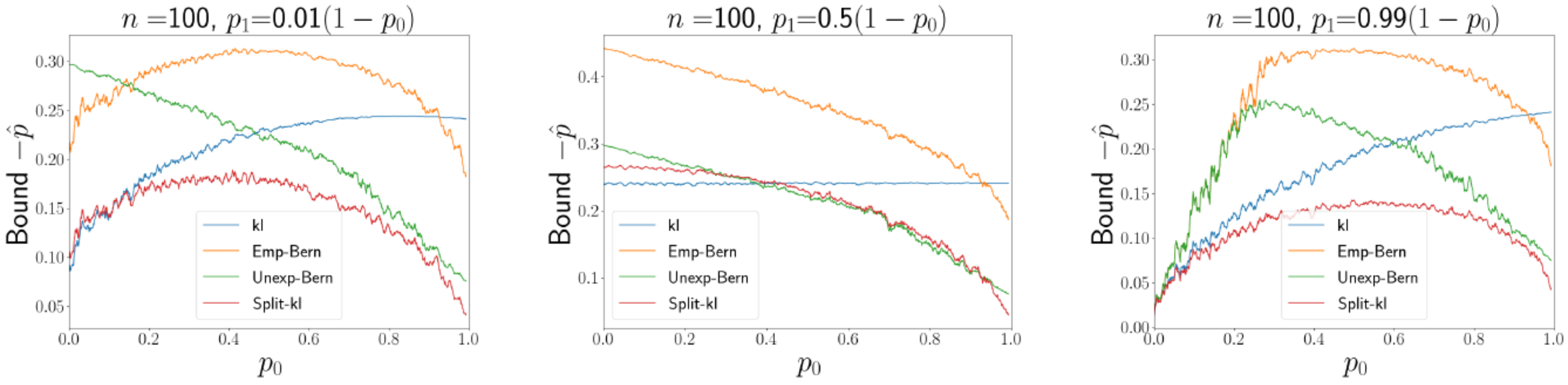
- Theorem: Let $X_1, \dots, X_n$ be i.i.d. with $X_1 \leq b$. Let $\mu = \mathbb{E}[X_1]$, $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n X_i$, $\hat{s}_n = \frac{1}{n}\sum_{i=1}^n X_i^2$. Let $\psi(u) = u - \ln(1+u)$. Then for any $\lambda \in \left[0, \frac{1}{b}\right)$:

$$\mathbb{P}\left(\mu \geq \hat{\mu}_n + \frac{\psi(-\lambda b)}{\lambda b^2}\hat{s}_n + \frac{\ln\frac{1}{\delta}}{\lambda n}\right) \leq \delta.$$

- Proof: Chernoff's bounding technique + Unexpected Bernstein's Lemma
- Directly relates $\mu - \hat{\mu}_n$ to $\hat{s}_n$ (empirical quantity) without going in two steps through ν (oracle quantity), as in Empirical Bernstein
- This time the optimal value of λ depends on $\hat{s}_n$. Therefore, we need to take a union bound over a set of values λ_j and pick the best.

Empirical Comparison

- kl, Empirical Bernstein, Unexpected Bernstein, split-kl
- Ternary r.v. in $\{-1,0,1\}$



(a) (b) (c)

Figure 4: Comparison of the concentration bounds with $n = 100$, $\delta = 0.05$, and (a) $p_1 = 0.01(1-p_0)$ and $p_{-1} = 0.99(1-p_0)$, (b) $p_{-1} = p_1 = 0.5(1-p_0)$, (c) $p_1 = 0.99(1-p_0)$ and $p_{-1} = 0.01(1-p_0)$.

Summary

- Bernstein: $\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\nu \ln\frac{1}{\delta}}{n}} + \frac{b \ln\frac{1}{\delta}}{3n}\right) \leq \delta$
  - Oracle bound in terms of $\mathbb{V}[X_1] \leq \nu$
- Empirical Bernstein: $\mathbb{P}\left(\mu \geq \hat{\mu}_n + \sqrt{\frac{2\hat{\nu}_n \ln\frac{2}{\delta}}{n}} + \frac{7 \ln\frac{2}{\delta}}{3n}\right) \leq \delta$
  - Empirical bound in terms of $\hat{\nu}_n = \frac{1}{n(n-1)}\sum_{1\leq i<j\leq n}\left(X_i - X_j\right)^2$
  - Bounding proceeds in two steps: bound ν in terms of $\hat{\nu}_n$ and plug into Bernstein
- Unexpected Bernstein: $\mathbb{P}\left(\mu \geq \hat{\mu}_n + \min_{\lambda\in\Lambda}\left(\frac{\psi(-\lambda b)}{\lambda b^2}\hat{s}_n + \frac{\ln\frac{|\Lambda|}{\delta}}{\lambda n}\right)\right) \leq \delta$
  - Empirical bound in terms of $\hat{s}_n = \frac{1}{n}\sum_{i=1}^{n} X_i^2$
  - Direct bound
  - Requires a grid Λ of λ values and a union bound over it